

CASE STUDY · RETAIL OPERATIONS

Reducing Stockouts Without Overstocking

A worked walkthrough of the classic inventory tension: running out of best-sellers while cash sits tied up in slow movers and how ABC analysis turns a 2,000-SKU problem into a manageable one.

The Scenario

A retail operation with roughly 2,000 SKUs is seeing two problems at once: customers reporting popular items are frequently out of stock, while the warehouse team reports storage costs climbing from inventory that isn't moving. Reordering is currently done the same way for every SKU a flat "reorder at 20% of max stock" rule.

Step 1 — Why a Flat Rule Fails

Treating 2,000 SKUs identically is the root cause. A product that sells 500 units a week and one that sells 2 units a month have completely different stockout costs and completely different carrying costs but the flat rule ignores that entirely. The fix isn't a smarter formula per SKU yet; it's first admitting not all SKUs deserve the same attention.

Step 2 — ABC Classification

Rank every SKU by revenue contribution, then bucket by cumulative share — this is the Pareto principle applied to inventory:

Class	% of SKUs	% of Revenue	Policy
A	~20%	~75-80%	Tight stock control, frequent review, never stock out
B	~30%	~15-20%	Standard review cycle, moderate buffer
C	~50%	~5%	Minimal stock, order-on-demand where possible

This single table is usually the turning point in a project like this it's the moment "2,000 SKUs" stops being an unmanageable number and becomes "400 SKUs that actually need daily attention."

Step 3 — Match the Stockout Problem to the Right Class

Pull the specific SKUs customers are complaining about and check which class they fall into. In a case like this, the pattern is usually telling, most stockout complaints cluster in Class A, while most of the excess warehouse inventory sits in Class C. The flat reorder rule was simultaneously under-protecting the products that mattered most and overstocking the ones that mattered least.

Step 4 — Right-Size the Reorder Logic

With classes defined, the reorder policy can finally differentiate:

- **Class A:** shorter review cycles, higher safety stock, prioritized in supplier negotiations
 - **Class B:** standard cycle, moderate buffer — no special treatment needed
 - **Class C:** consider a simpler policy altogether, like order-on-demand or consolidated quarterly orders, since the carrying cost of precision here outweighs the benefit
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Step 5 — Watch for the Trap: Revenue Isn't the Only Axis

A pure revenue-based ABC split can miss SKUs that are low-revenue but strategically important. A loss-leader that drives footfall or a product tied to a key account. A more complete version of this analysis cross-tabs revenue class against a second dimension (margin, or strategic flag) before finalizing policy. Worth flagging explicitly in any real write-up, so the recommendation doesn't get blindly automated.

What This Case Study Teaches

ABC analysis is simple to compute but easy to apply carelessly. Its real value isn't the classification itself it's that it forces a business to stop spending equal attention on unequal problems.

This is a practice walkthrough for learning analytical structure — not a record of an actual client engagement. Terence Rebello · The Analyst's Toolbox